Vision-Language Models for Interpretable Chest X-ray diagnosis

University of Exeter

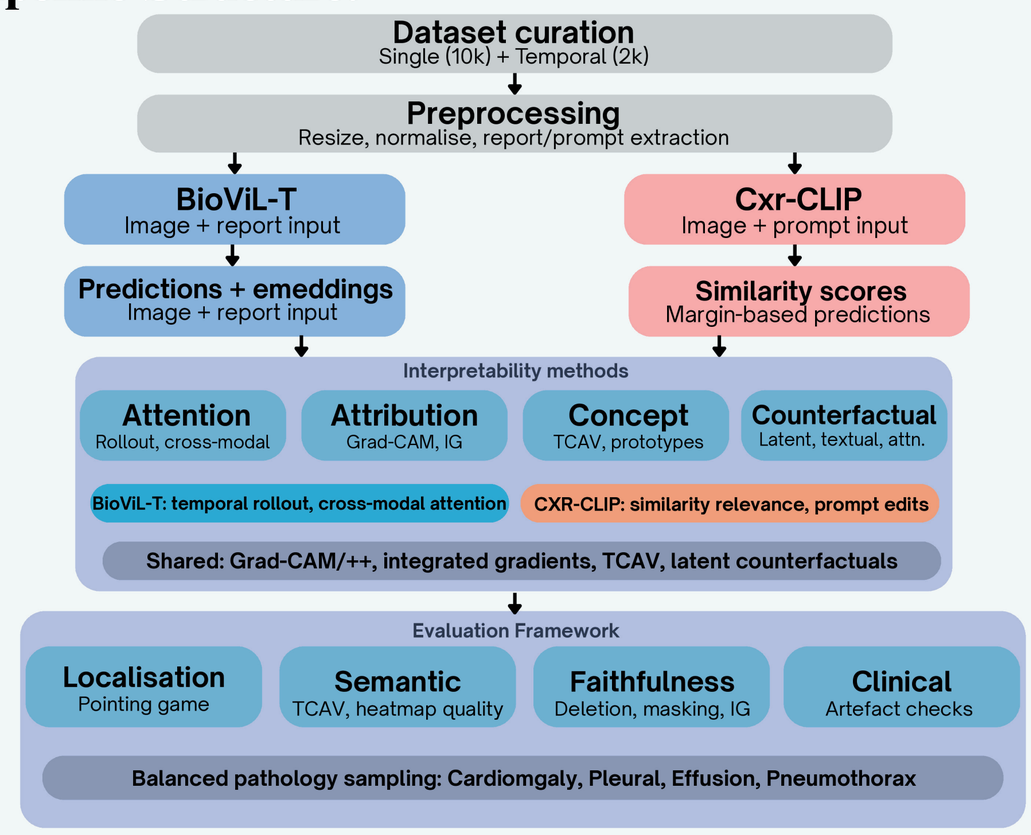
Problem Statement:

Vision-Language models have recently emerged as promising tools for automated medical imaging diagnostics. However, these models often function as black boxes, providing limited explanation for their predictions.

Aims & Objectives:

- Curate smaller subsets of the MIMIC-CXR dataset.
- Implement a variety of interpretability techniques.
- evaluate and compare these techniques by analysing visualization and computing the appropriate metrics

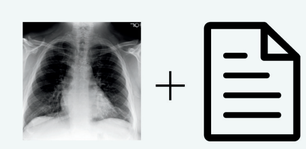
Pipeline Structure:

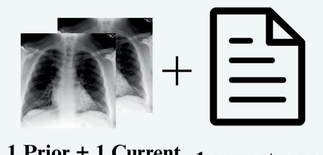


This unified inference pipeline produces embeddings and Similarity scores which were then analysed using attention, attribution, concept-based and counterfactual interpretability and, then evaluated using loclaisation, semantic alignment, faithfulness and, clinical plausibility.

Dataset Curation:

- The MIMIC-CXR-JPG dataset was split into two subsets, Single (10,000 Samples) and Temporal (2000 Samples).
- Samples were de-duplicated and prioritized based on specific diseases, View positions, image quality and report length using a point system.



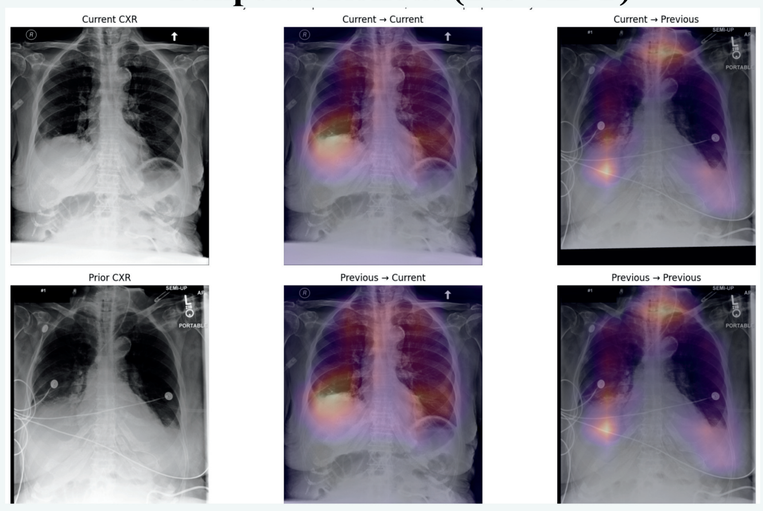


Interpretability Heatmaps:

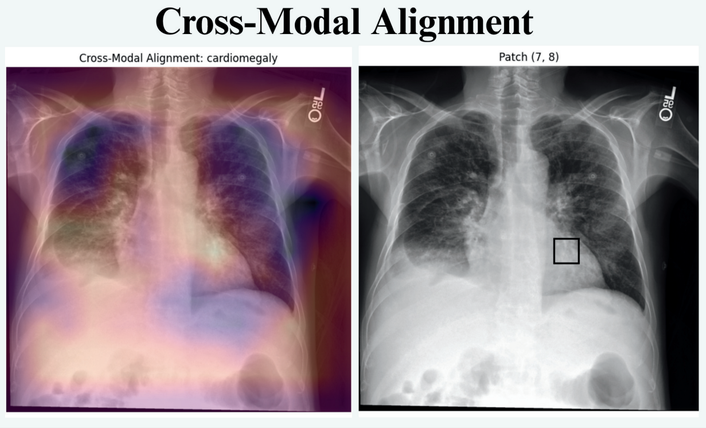
Spatial-Rollout (BioViL-T)



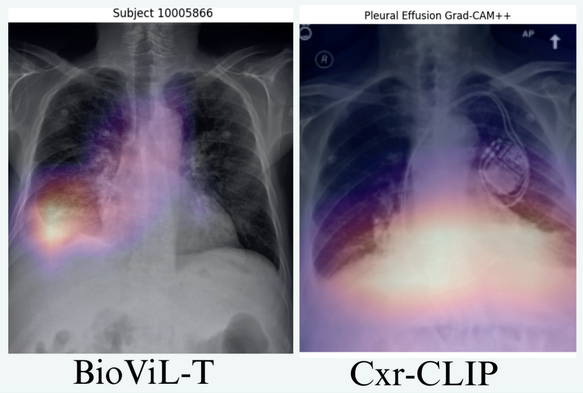
Temporal-Rollout (BioViL-T)



Cross-Modal Alignment

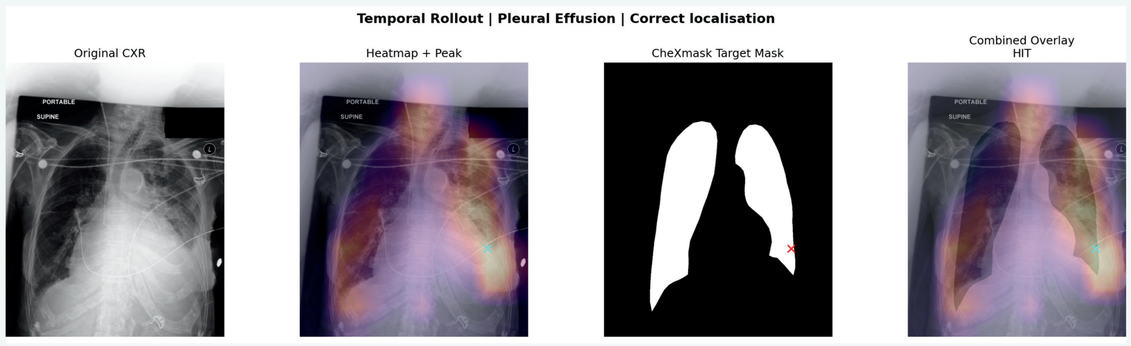


Grad-CAM / Grad-CAM++



Evaluation Results

Localisation Quality (LQ)



BioVil-T Spatial Rollout (Balanced) — Pointing Game

Pathology	N	Hits	Score
Cardiomegaly	199	49	0.2462
Pneumothorax	147	33	0.2245
Pleural Effusion	539	33	0.0612
Overall	885	115	0.1299

BioViL-T Temporal Rollout (Balanced) — Pointing Game

Pathology	N	Hits	Score
Cardiomegaly	199	20	0.1005
Pneumothorax	147	18	0.1224
Pleural Effusion	539	52	0.0965
Overall	885	90	0.1017

BioViL-T Grad-CAM / Grad-CAM++ – Pointing Game

Pathology	N	Hits	Score
Cardiomegaly	284	88	0.3099
Pneumothorax	69	16	0.2319
Pleural Effusion	341	11	0.0323
Overall	694	115	0.1657

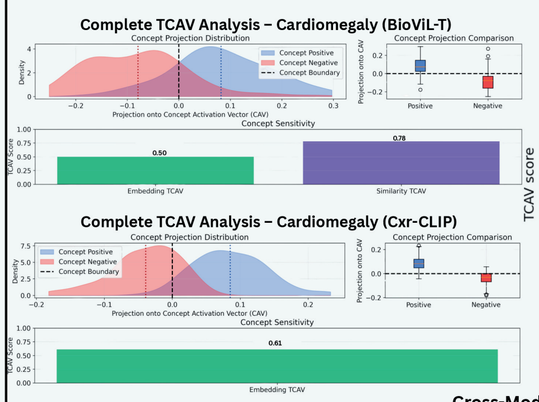
Cxr-CLIP Grad-CAM / Grad-CAM++ – Pointing Game

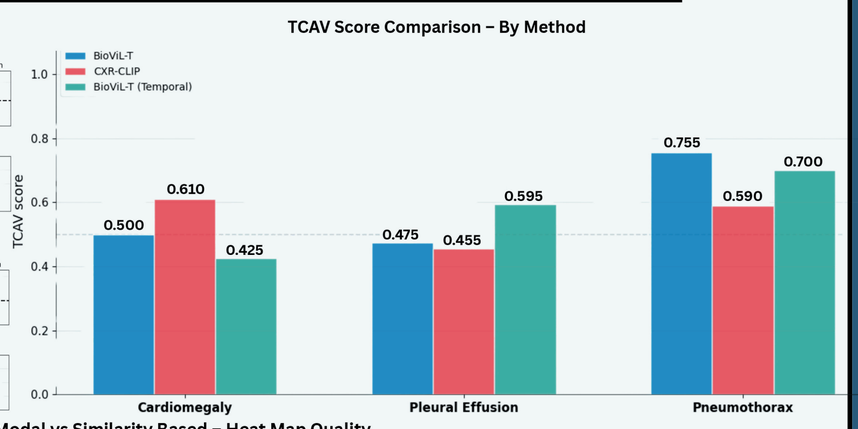
Pathology	N	Hits	Score
Cardiomegaly	284	231	0.8134
Pneumothorax	69	5	0.0725
Pleural Effusion	341	7	0.0205
Overall	694	243	0.3501

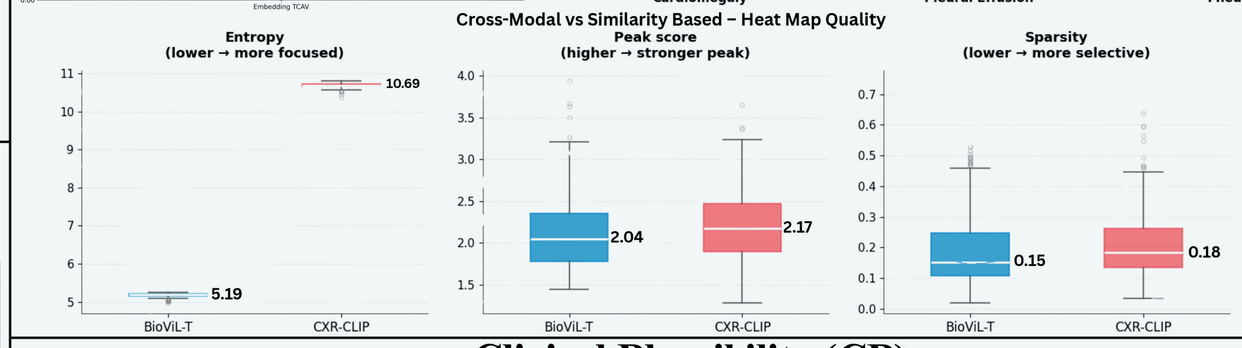
Semantic Alignment (SA)

Semantic Alignment – Complete Summary

Metric	BioViL-T	Cxr-CLIP	Notes
Embedding TCAV (mean)	0.5767	0.5517	Higher= more concept sensitive
Similarity TCAV (mean)	0.6450	N/A	BioViL-T only (paired reports)
Heatmap Entropy (mean)	5.1757	10.6788	Lower = more focused
Peak Score (mean)	2.1241	2.1976	Higher = stronger localisation
Sparsity (mean)	0.1903	0.2133	Lower = more selective



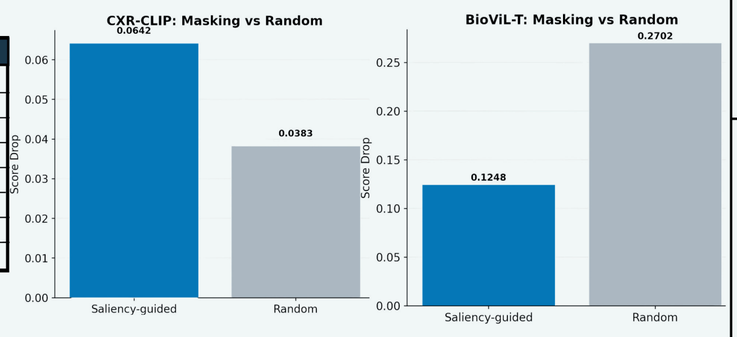


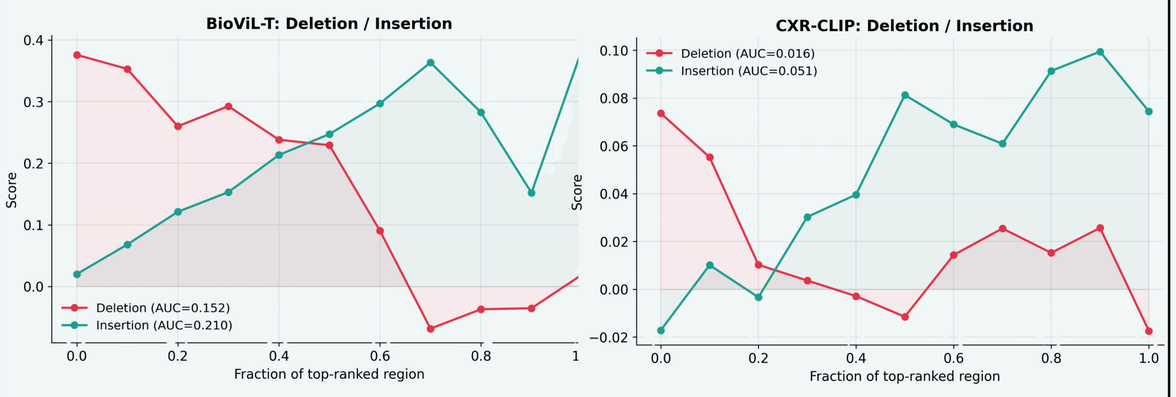


Faithfulness (F)

Faithfulness Comparison: BioViL-T vs. Cxr-CLIP

Metric	BioViL-T	Cxr-CLIP
Deletion AUC	0.0281	-0.0271
Insertion AUC	0.4163	0.1108
Saliency Mask Drop	0.6023	0.0757
Random Mask Drop	0.8101	0.0478
Saliency Blur Drop	0.4989	0.0257
Random Blur Drop	0.0678	0.0008
Mask > Random %	0.0667	0.6000
Blur > Random %	0.08667	0.4667





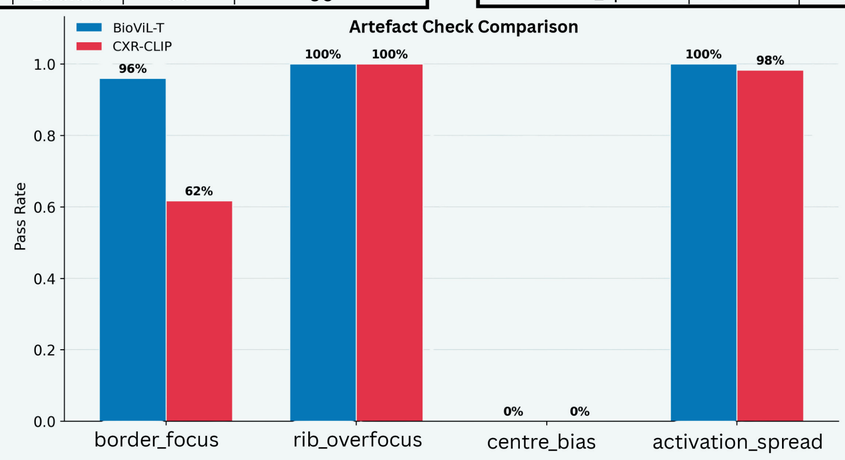
Clinical Plausibility (CP)

BioViL-T: Artefact Check Summary

Check	Pass Rate	Passed	Total
border_focus	96.0%	48	50
rib_overfocus	100.0%	50	50
centre_bias	0.0%	0	50
activation_spread	100.0%	50	50

Cxr-CLIP: Artefact Check Summary

Check	Pass Rate	Passed	Total
border_focus	61.7%	37	60
rib_overfocus	100.0%	60	60
centre_bias	0.0%	0	60
activation_spread	98.3%	59	60



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